Swedish Metabarcoding Network

Mobilising eDNA data: new free services to make your data FAIR

eDNA journal club at EBC · Friday 23 October 2026 · 15:00–16:00

Homo sapiens room · Tobias Andermann · Linus Lassen



TODAY

What we will cover

1

Who we are

The network, its mission and goals

2

Why mobilise eDNA data

FAIR requirements and what you gain

3

How: Live – Demonstration for ASV portal upload

The route from ASV tables to published data

4

Get involved

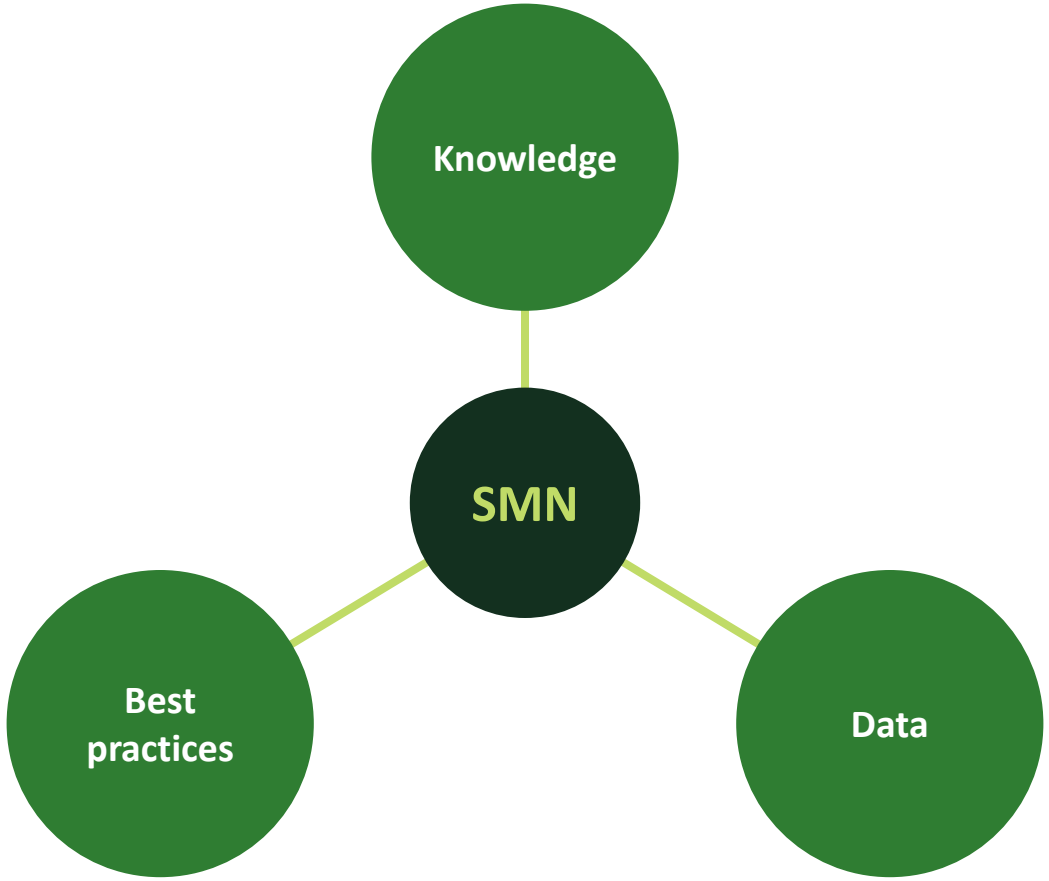
Free help, training and discussion

Connecting knowledge, best practices and data

SMN aims to improve the technology readiness of metabarcoding as a tool for effective biodiversity monitoring.

We are a network of partners across Sweden who develop and share best practices, protocols and data – from fieldwork via labwork to data analysis and publication.

Particular emphasis: making existing and newly produced datasets available as FAIR biodiversity data.



WHO WE ARE

Our goals

1

Improve the technology readiness of metabarcoding for effective biodiversity monitoring

2

Create a network of researchers and biodiversity practitioners across Sweden

3

Enable researchers and non-academic partners to utilise metabarcoding data

4

Advance FAIR biodiversity data by mobilising metabarcoding data to the ASV Portal

Goal 4 is the focus of today's session.

From sample to FAIR biodiversity data



1 Field sampling

2 DNA extraction

3 PCR & library prep

4 Sequencing

5 ENA

6 Bioinformatics

7 ASV Portal

8 FAIR data

SMN shares protocols and tutorials for every step – today we focus on **steps 5–8: getting your data published and reusable.**

Your data deserve a second life

Today

- Many eDNA datasets end up only in supplementary files or on local drives
- Raw reads in ENA/SRA are rarely reusable as species occurrences
- Taxonomic assignments go out of date as reference databases improve

Increasingly expected

- Funding agencies increasingly require FAIR, openly shared data
- Data management plans must say where data will be published

Your data deserve a second life

Vetenskapsrådet

På svenska

Start / Applying for funding / Requirements, terms and conditions / Plan, manage and make research data available

Plan, manage and make research data available

If you are awarded a grant from the Swedish Research Council, you must manage your research data in accordance with the FAIR principles and make the data available according to the principle of "as open as possible, as restricted as necessary". You must also produce, maintain and follow a data management plan. The plan should describe how research data will be managed during the course of the research and how they will be handled afterwards.

What you gain from the ASV portal

1 Visibility

A permanent GBIF DOI makes each dataset citable and traceable

2 FAIR data

Consistent metadata, open formats and long-term accessibility

3 Up to date

ASVs are regularly re-annotated with the latest reference databases

4 Insight & comparison

Compare your results with other metabarcoding datasets

5 Global knowledge

Data flow to GBIF and support biodiversity research worldwide

6 Environmental context

Explore data alongside environmental layers in the spatial portal

HOW IT WORKS

From raw reads to published biodiversity data



Free help along the whole route: SMN data stewards can assist with ENA submission, metadata and ASV/OTU upload.

Alternative for OTU/BIOM data: GBIF Metabarcoding Data Toolkit (MDT).

LIVE DEMO

Demo

From ASV table to published biodiversity data

ENA → **ASV portal** → **SBDI Bioatlas** → **GBIF**

raw reads & sample metadata

upload template

searchable occurrences

global & citable

HOW IT WORKS

Recipe for ASV portal upload

- ✓ ASV table with read counts per sample
- ✓ ASV sequences and taxonomic assignments
- ✓ Sample metadata: date, coordinates, sampling protocol
- ✓ ENA accessions for samples and sequence runs
- ✓ Methods: target gene & region, primers, sequencing platform, denoising tool

Already submitted to ENA?
Then you have most of the sample metadata ready, and your ENA accessions link the two records.

Data Support from NBIS

Contact person: Britt Andermann

NEW SERVICE

eDNA lab work now available at NGI



Lobbied for by SMN: started as a pilot with Planetary Biology, now available through NGI (official launch to come).

The whole lab workflow for eDNA metabarcoding in one place, a big help for our community.

GET INVOLVED

Join the network

1. **Sign up for the newsletter**
   Events, trainings, resources and opportunities – free, no obligations

2. **Fill in the survey**
   If you produce metabarcoding data and want to make it FAIR

3. **Request help from a data steward**
   Free support for uploading your ASV/OTU data

4. **Access guidelines, best practices**
   Positions, calls or collaborations – we post them on the website

metabarcoding.se

smn-members@lists.uu.se

GET INVOLVED

Join the first in-person event hosted by SMN!

When: 8th to 9th of December

Where: Uppsala

What: eDNA Workshops, discussions and more

Stay tuned, updates will follow on our webpage, newletter list and journal club mail

metabarcoding.se

DISCUSSION

Let's talk about your eDNA data

1. What eDNA / metabarcoding data do you have that is not yet published?

2. What stops you from sharing it – time, metadata, formats, permissions?

3. Which support would help you most: data stewards, tutorials, workshops?



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Thank you!